



Stratigraphic categorisation as documentation: the Extended Matrix and the data/taxonomy plane of credible virtual reconstruction

*A worked example from the EM 1.5 specification, developed within the
Horizon Europe StratiGraph project as a building block for the European
Collaborative Cloud for Cultural Heritage (ECCCH).*

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Abstract

Research questions and objectives. Best documentation practices for virtual reconstructions of cultural heritage are increasingly demanded by international frameworks – the London Charter (Denard, 2009), the Principles of Seville (López-Menchero & Grande, 2011), and the forthcoming Vienna Charter now in preparation. This paper addresses a question often left implicit: what epistemic operation must be in place before documentation can occur? The hypothesis is that documentation presupposes categorisation – the assignment of a stratigraphic identity to every modelled element, the distinction between findings and sources, and the declaration of temporal scope – and that this categorical layer constitutes a distinct data/taxonomy plane of virtual reconstruction documentation, upstream of the measurement of uncertainty and of dissemination.

Context and related work. The contribution builds on the chartered tradition of London (Denard, 2009) and Seville (López-Menchero & Grande, 2011), on the semantic-interoperability foundation of CIDOC CRM (Doerr, 2003), on the uncertainty-modelling tradition centred on the scientific reference model (Apollonio, 2016; Kuroczyński, Apollonio, Bajena & Cazzaro, 2023), and on the documentary-infrastructure tradition of IDO VIR (Wacker, Stille, Grellert, Bruschke & Beck, 2022). Its deeper root, however, lies in twentieth-century restoration theory: the theory of lacuna (Brandi, 1963), by which reconstructed portions of a heritage object must remain visually and semantically distinguishable from attested fabric. The Extended Matrix (EM) – a stratigraphic knowledge graph aligned to CIDOC CRM (Demetrescu, 2015; Demetrescu & Fanini, 2017) – operationalises this

principle in digital space: it represents each reconstruction as a graph of typed stratigraphic units that explicitly declare which portions are attested, which are reconstructed from documentary evidence, and which are interpretive hypotheses.

Methods and data. The current EM 1.5 specification – developed within the Horizon Europe StratiGraph project as a building block for the European Collaborative Cloud for Cultural Heritage (ECCCH) – consolidates a stratigraphic node typology (US, USD, USVs, USVn, SF, VSF), in which the Documentary Stratigraphic Unit (USD) represents elements attested by documentary sources but no longer physically preserved. The specification adopts an existence-based colour scale grounded in reproducible taxonomic principles: the categorisation of each stratigraphic unit follows stable epistemic criteria, independent of the contingent configuration of the available sources – which, in favourable and not infrequent cases, may grow over time without altering the categorisation itself (Figure 1). The graph substrate is navigable through Heriverse, a web application under active development within the StratiGraph project, which visualises the EM graph directly within the 3D space of the virtual reconstruction.

Expected and achieved results. The combination of EM 1.5 and Heriverse yields a documentation substrate that is simultaneously machine-actionable (graph-queryable, FAIR-aligned, ECCCH-ready) and human-legible (time-oriented knowledge graph and colour-coded 3D representation). The presentation will include visual examples of navigating an EM 1.5 graph in a virtual reconstruction. The paper also maps the data/taxonomy plane onto a broader three-plane reading of virtual reconstruction documentation – categorisation (data/taxonomy), measurement of uncertainty, and process-documentation and dissemination.

Relevance to the conference themes. The contribution falls primarily within Theme 2 (Best Documentation Practices) – addressing knowledge-graph and semantic-web approaches, ontology-aligned vocabularies, FAIR principles, and the documentation of hypothetical historical research – and cuts across Theme 1 (3D Web Exploration and Visualisation), through Heriverse as a viewer integrating knowledge graphs with 3D space. The aim is to contribute to the forthcoming Vienna Charter a stratigraphic-archaeological complement to the architectural and dissemination-oriented frameworks currently shaping the discussion.

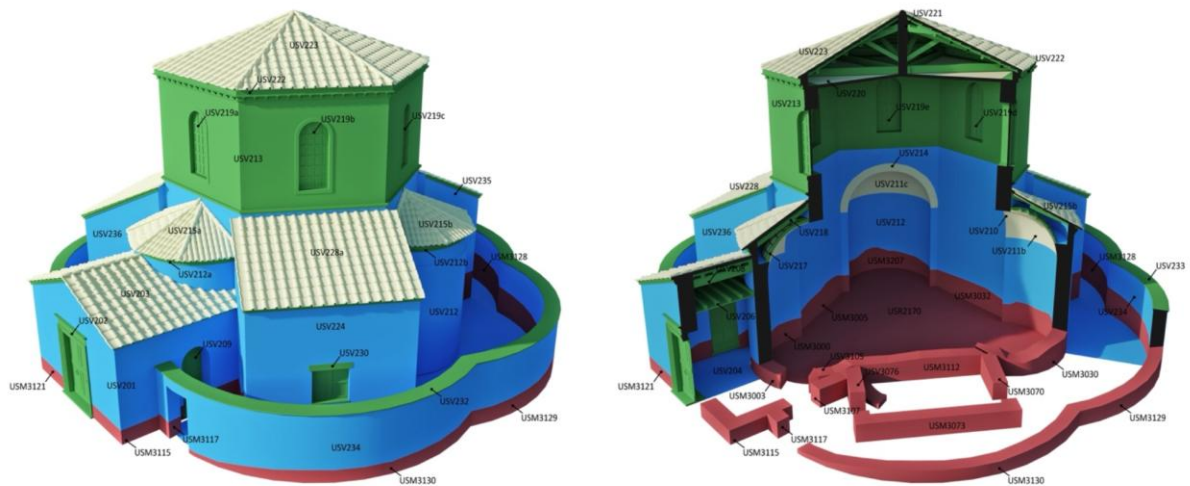


Fig. 1 Worked-example case study — Late Roman villa of Aiano (Italy). The 3D reconstruction integrates stratigraphic units (US, USVs, USVn) typed and colour-coded according to the EM specification (cf. Figure 2): red for attested stratigraphic units (USM, US), blue for structural virtual reconstructions (USVs), green for non-structural virtual reconstructions (USVn). After Ferdani et al. (2020), Fig. 10, p. 15.

Extended Matrix — node colour scale (valid from EM 1.4+)

Node	Definition	Description	Col	R	G	B	Hex
US	Extant structures	In situ archaeological evidences		0.328	0.033	0.033	9B3333
USD	Documented structures	Attested by historical sources, not preserved		0.549	0.103	0.000	D86400
USV/s	Structural virtual reconstruction	Based on in situ archaeological evidences		0.018	0.275	0.799	248FE7
USV/n	Non structural virtual reconstruction	Based on sources and comparisons		0.031	0.191	0.026	31792D
SF	Fragmental virtual anastylosis	Repositioning of non in situ architectural fragments		0.847	0.741	0.188	D8BD30
VSF	Fragmental virtual reconstruction	Completion of repositioned fragments		0.694	0.623	0.380	B19F61

extendedmatrix.org a

Fig. 2 Extended Matrix 1.5 colour schema

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